

Aaron Dutta

647-244-5263 | aaron.dutta@uwaterloo.ca | [in aaron-dutta](https://www.linkedin.com/in/aaron-dutta) | [defAaron](https://github.com/defAaron) | aaron.dutta.com

EDUCATION

University of Waterloo

BMath, Honours Mathematics with Co-Op

Sep 2026 - Present

Waterloo, ON

SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL

Libraries/Frameworks: React.js, Next.js, Node.js, FastAPI, RestAPIs, Flask, Hono, Prisma

AI/ML & Data: OpenCV, MediaPipe, NumPy, Redis, Pandas, ONNX Runtime, Pydantic

Tools & Platforms: Git, Docker, PostgreSQL, Azure, Cloudflare, GCP, Vercel, Supabase, Git

EXPERIENCE

Software Engineering Intern

September 2026 – Present

Magic Hour (YC W24)

San Francisco, CA

- Designing and shipping API-integrated products that showcase Magic Hour's video-generation capabilities and grow adoption of the public developer API
- Building full-stack demos and tooling with **Hono**, **PostgreSQL**, **Redis**, and **Cloudflare R2/S3**, using **ffmpeg** for transcoding and media assembly
- Accelerating AI-backed features with **Typesafe AI Jev** and **Meta Muse** while keeping production-oriented data and storage patterns

Executive Design Engineer

August 2026 – Present

UW Cube

Waterloo, ON

- Building and maintaining the Cube public website in **Next.js** and **TypeScript** and integrating an AI chatbot by connecting **Python/LangChain** backend services
- Leading **GitHub** collaboration as an executive design engineer, reviewing PRs with software engineers and aligning design and implementation within a cross-functional team
- Designing internal tools to streamline developer workflows, reducing manual steps during building and shipping

PROJECTS

Indago | *Python, dbt, DuckDB, Polars, Anthropic API, LangGraph, Pydantic* | [🔗](https://github.com/defAaron/indago) /

Sep 2026 – Present

- Built a graph-grounded root-cause agent for **dbt** data quality incidents that delivers cited diagnoses to reduce manual upstream tracing for data engineers
- Engineered **dbt** and **DuckDB** pipelines on NYC taxi and synthetic e-commerce data with staged models, automated tests, and archived manifests per run
- Implemented deterministic origin-node localization using a typed lineage graph, **Polars** drift checks, and git history
- Shipped **LangGraph** on Anthropic's Claude models with retrieval tools and fault-injection eval (**Pydantic** and **147-pytest suite**)

SkyEye | *Python, Flask, YOLOv8, ONNX Runtime, React, TypeScript* | [🔗](https://github.com/defAaron/skyeye) [🔗](https://github.com/defAaron/skyeye)

Aug 2026

- Won RescueHacks 2026 for an SAR triage tool that turns a free-text missing-person report into a Lost Person Behavior search ring, scoring map pins in **5–8s** within a **30s** target
- Developed a **sliding-window YOLOv8n + ONNX Runtime** pipeline to tile images into overlapping windows
- Shipped a **Flask + React/TypeScript** operator console with Gemini Flash (Groq fallback) for typed extraction, and **GCP** for geocoding locations and drawing search zones

AWARDS

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|---|----------|
| • Winner @ RescueHacks, 160+ hackers | Sep 2026 |
| • 2nd Place Overall @ Katy Youth Hacks, 220+ hackers | Aug 2026 |
| • 3rd Place Security/Privacy Hack @ STEMist Hacks, 330+ hackers | Aug 2026 |
| • 1st Place @ STEMMINDS Hackathon, 60+ hackers | Mar 2026 |